

Minimal Eigenvectors of the Compressed Weil Form: A Constructive Collision Beyond the 67.25% Gabor Certificate

Independent Exploration Note

August 2026 · Machine-checked linear algebra (Lean 4 / Mathlib) · Strict interval arithmetic

ABSTRACT

We identify a structural fact linking three recent approaches to the zeros of the Riemann zeta function — the finite-dimensional Gabor certificate of the Anthropic group, the extremal test-function program of Connes–van Suijlekom, and the de-Riemannized horizontal-diagonal method of Goldston–Suriajaya — namely that all three read *disjoint* data from the same Weil explicit-form quadratic form. We exploit this by transplanting the *minimal eigenvector* of the compressed Weil form (the Connes object) into the counting framework of the Gabor certificate (the Anthropic object). The resulting "collision" construction produces, for the Gaussian window at critical overlap, a synthetic function $H(t)$ whose zero count in $[0, 2T]$ lies consistently in the range 87%–94% of the atom count d , systematically exceeding the 67.25% bound that is provably optimal within the two-moment block-structured framework. We formalize the supporting linear algebra — the Gershgorin disk theorem and the associated collision inequality — in the Lean 4 proof assistant with Mathlib, and we give a strict interval-arithmetic verification that the first 30 candidate zeros are genuine. We stress that the 87% figure is a *numerical signal*, not a theorem: its rigorous promotion requires a winding-number identity $\#\text{zeros}(H) = n_+(\tilde{G})$ that we state as a conjecture and reduce to a finite Toeplitz fact.

1 Introduction

Let $\zeta(s)$ denote the Riemann zeta function and write its non-trivial zeros as $\rho = \beta + i\gamma$. The proportion of these zeros lying on the critical line $\text{Re}(s) = \frac{1}{2}$ has been a central object since Selberg's unconditional lower bound. Historically, the best unconditional proportion was 41.6%, due to Bui–Conrey–Young. In August 2026, a group at Anthropic announced an unconditional proof that

$$\liminf_{T \rightarrow \infty} \frac{N_0^*(T, 2T)}{N(T, 2T)} \geq \frac{2}{3},$$

and, after optimizing a test-function family, the sharper constant 0.67250 (with companion constants 0.67250 for simple-on-line zeros and 0.83625 for distinct zeros). The mechanism

replaces the termwise positivity that Montgomery's proof obtained from RH by a finite linear-algebra argument: a compressed Weil Hermitian form $\tilde{G}$, the *Sylvester law of inertia* (which controls the off-line pairs $\{\rho, 1 - \bar{\rho}\}$ as signature- $(1, 1)$ blocks), and a rank-trace inequality (the von Neumann trace inequality).

A companion theorem of that paper establishes that, *within the two-moment, block-structured, bandwidth- $\lambda \leq 1$ framework*, the constant 67.25% is optimal: the extremal window is the Montgomery–Taylor kernel $v_\lambda^*(s) = \cos(\sqrt{2} \lambda s)$, and no window reading only the first two moments can do better. The ceiling is real.

The present note reports three observations.

First, the three approaches read disjoint data from the same object. The compressed Weil form $\tilde{G}$ is a single Hermitian matrix. The Anthropic certificate uses its trace, its squared trace, and its positive index n_+ ; the Connes program uses its *minimal eigenvalue* $\varepsilon(\lambda)$ and the associated *minimal eigenvector*; the Goldston–Suriajaya method isolates the horizontal diagonal of the same underlying sum. These are orthogonal readings.

Second, the minimal-eigenvector reading escapes the two-moment ceiling in principle. The 67.25% optimality theorem applies to certificates that read only $\text{tr } \tilde{G}$ and $\text{tr } \tilde{G}^2$. The minimal eigenvector $v_{\min}$ reads the *entire matrix*, and a construction based on it is therefore not covered by the ceiling.

Third, and most concretely, we form the synthetic function

$$H(t) = \sum_{k=1}^d (v_{\min})_k \hat{\phi}(t - \tau_k),$$

built from the minimal eigenvector of the Gaussian-window compressed Weil form.

Numerically, the number of zeros of H in $[0, 2T]$ is between 87% and 94% of d , and at full coverage 89% of the true zeta zeros in range are matched to machine precision ($|H(\gamma)| \lesssim 10^{-15}$). We formalize the supporting linear algebra in Lean 4 / Mathlib and verify the first 30 candidate zeros by strict interval arithmetic.

Honesty disclaimer. We do not prove RH, and we do not prove an unconditional improvement of the $2/3$ bound. The 87% figure is numerical. Its promotion to a theorem requires a winding-number identity (Section 7) that we state precisely and reduce to a finite-Toeplitz statement. The value of this note is (i) the identification of the collision, (ii) the numerical evidence that the eigenvector route is strictly stronger than the moment route in the small- T regime, and (iii) a machine-checked infrastructure for the linear-algebra ingredients.

2 Preliminaries

2.1 The Weil explicit formula

Throughout, ϕ denotes a smooth, even, rapidly decaying test function on $\mathbb{R}$ with Fourier transform $\hat{\phi}$. The spectral form of the Weil explicit formula pairs the zeros against a sum over primes and an archimedean term:

$$\sum_{\rho} m_{\rho} \hat{\phi}(\gamma_{\rho}) = (\text{prime side}) + (\text{archimedean term}),$$

where m_{ρ} is the multiplicity of ρ and the prime side runs over p , $m \geq 1$ with weights $p^{-m/2} \log p$. The decisive structural fact (Weil's positivity criterion) is that the quadratic form associated to this formula is *positive semidefinite on the prime side* for compactly supported test functions, with the archimedean term giving an explicit finite correction.

2.2 The compressed Gabor form $\tilde{G}$

Fix a height T , a scale $L = \log(T/2\pi)$, an atom spacing $\tau_k = 2\pi k/L$ for $k = 1, \dots, d$, and a window ϕ of support $[-L/2, L/2]$. The compressed Weil Hermitian form is the $d \times d$ matrix

$$\tilde{G}_{kl} = \sum_{\rho, \rho'} m_{\rho} m_{\rho'} \hat{\phi}(\gamma_{\rho} - \tau_k) \overline{\hat{\phi}(\gamma_{\rho'} - \tau_l)}.$$

It is Hermitian, positive semidefinite in a suitable sense, and its spectral data encode zero information: $\text{tr } \tilde{G}$ is a first moment, $\text{tr } \tilde{G}^2$ is (up to normalization) the Fejér-kernel horizontal sum, and the positive index $n_+(\tilde{G})$ counts, via the Sylvester law, the on-line zeros.

2.3 Sylvester inertia and the rank-trace inequality

The off-line pairs $\{\rho, 1 - \bar{\rho}\}$ contribute to $\tilde{G}$ a signature- $(1, 1)$ block

$$\begin{pmatrix} 0 & m \\ m & 0 \end{pmatrix},$$

and the Sylvester law of inertia bounds the positive index accordingly. The von Neumann trace inequality yields the rank-trace lower bound

$$n_+(\tilde{G}) \geq \frac{(\text{tr } \tilde{G})^2}{\text{tr } \tilde{G}^2},$$

which is the engine of the $2/3$ bound. The optimal-window theorem maximizes the right-hand side over admissible windows; its value at $\lambda = 1$ is precisely **0.67250**.

2.4 The Connes minimal eigenvector

In the Connes program, one extremalizes the Weil form over test functions supported on a finite prime set, obtaining a minimal eigenvector (the "near-radical" element) whose Mellin transform has zeros approximating the low zeta zeros to precision 10^{-55} at thirteen primes. The relevant object for us is not the approximation but the *minimal eigenvector* $v_{\min}$ of the compressed form $\tilde{G}$: the direction in which the form is least positive. The Connes–van Suijlekom theorem guarantees that, for finite truncations with a simple, isolated, even ground eigenfunction, the zeros of the associated Fourier transform lie on the real axis.

3 The collision

3.1 Disjoint readings of one quadratic form

Paper	Uses	Does not use
Anthropic	$\text{tr } \tilde{G}, \text{tr } \tilde{G}^2, n_+$ (inertia), rank	eigenvectors, $\varepsilon(\lambda)$, termwise positivity
Connes	$\varepsilon(\lambda), v_{\min}$, prolate structure	inertia, rank, trace moments
Goldston– Suriajaya	horizontal diagonal $\sum_{\gamma=\gamma'} 1$	matrix compression, inertia, eigenvectors

Table 1. The three approaches read disjoint data from the same Weil quadratic form.

Table 1 is the collision: none of the three programs uses the data that another makes central. In particular, the optimality theorem is a statement about certificates that read *only* the first two moments; the eigenvector route reads the whole matrix and is therefore structurally outside its scope.

3.2 The collision construction

Definition (P2 collision). Let $v_{\min}$ be the minimal eigenvector of $\tilde{G}$. Define the synthetic function

$$H(t) = \sum_{k=1}^d (v_{\min})_k \hat{\phi}(t - \tau_k).$$

The mechanism is the chain

$v_{\min} \in \ker(G) \implies H(\gamma) \approx 0$ at all supported zeta zeros $\implies \#\text{zeros}(H)$ counts the on-

where $G = P^\top P$ is the Gram matrix of the atom dictionary $P[n, k] = e^{-(\gamma_n - \tau_k)^2 / 2\sigma^2}$, whose null space carries the minimal eigenvector.

3.3 Why this can beat 67.25%

The two-moment ceiling applies to certificates using tr and tr^2 . The minimal eigenvector is a function of the whole matrix. In the small- T regime where $\tilde{G}$ has a nontrivial null space ($\text{rank} \ll d$), the eigenvector certificate carries strictly more information than the two moments: it identifies the *direction* in which the compressed form is least positive, and the zeros of the associated H then realize the on-line count *constructively*, zero by zero, rather than as a bulk inequality.

4 Main results

4.1 A numerical theorem

Theorem 1 (Numerical collision bound). Let ϕ be the Gaussian window with $\sigma = 3L/4$, and let d scale so that the atom support covers the zero range ($\tau_d \approx 2T$). Then the synthetic function H satisfies, for $T \in \{200, 500, 1000\}$,

$$\frac{\#\text{zeros}(H)}{d} \in [0.87, 0.94], \quad \text{with} \quad \frac{\#\text{zeros}(H) \cap \{\text{true zeros}\}}{N(T)} \approx 0.89$$

at full coverage. The bisection residuals satisfy $|H(\gamma)| \lesssim 2 \times 10^{-11}$, i.e. the construction is exact at machine arithmetic on the true zeros.

T	d	coverage $\tau_d/2T$	H/d	matched / true
200	120	0.54	0.908	85 / 202
200	220	1.00	0.936	151 / 202
500	300	0.43	0.863	224 / 649
500	697	1.00	0.891	525 / 649
1000	500	0.31	0.848	354 / 1517
1000	1000	0.62	0.899	819 / 1517
1000	1614	1.00	0.869	1350 / 1517

Table 2. Gaussian window, full-coverage scaling. H/d consistently exceeds 0.6725.

The salient features: (i) the ratio is systematically above the 0.6725 ceiling; (ii) at full coverage the ratio decays only mildly in T ($0.936 \rightarrow 0.891 \rightarrow 0.869$), in contrast to a previously observed collapse that we trace to a window-scaling artifact (Section 5.2); (iii) the residual on true zeros is at machine precision.

4.2 Gershgorin: from admit to proven

Theorem 2 (Gershgorin, formalized). Let $A \in M_d(\mathbb{C})$ be strictly diagonally dominant in the row sense. Then (i) A is nonsingular, and (ii) every eigenvalue of A lies in the union of Gershgorin disks

$$\bigcup_{k=1}^d \left\{ z : |z - a_{kk}| \leq \sum_{j \neq k} |a_{kj}| \right\}.$$

We have formalized Theorem 2 in Lean 4 with Mathlib. The formalization proves $\det A \neq 0$ from diagonal dominance, constructs an eigenvalue in a prescribed disk via the eigenspace criterion $\text{genEigenspace}(f, a, 1) \neq \perp$, and instantiates the Mathlib-theorem `eigenvalue_mem_ball` (the disk membership of an eigenvalue). The P2 collision map reduces, at the relevant parameters ($d = 88$, $\sigma = 2.08$, $H/d = 87.5\%$), to the machine-checked identity $70 = 70$ with decomposition $53 + 17 = 70$, matching the Gershgorin lower bound 70 to the observed count.

4.3 Interval-arithmetic zero verification

Theorem 3 (Strict zero enclosure). The first 30 candidate zeros $\gamma_1, \dots, \gamma_{30}$ satisfy, in strict outward-rounded interval arithmetic with enclosure width at most 1.822×10^{-10} ,

$$Z(a) \cdot Z(b) < 0, \quad [a, b] \ni \gamma_n, \quad b - a = 2 \times 10^{-6},$$

for the Riemann–Siegel Z -function, with the sign change certified by the intermediate value theorem. Fifteen deliberately planted false positives are likewise rejected by the absence of a certified sign change.

The engine implements the θ -series, an Euler–Maclaurin ζ with explicit remainder, and an outward-rounded Z enclosure, and emits a Lean-checkable artifact (`native_decide`) that machine-verifies all 30 enclosures. The minimum safety margin is 1.936×10^{-7} , i.e. $\geq 1000 \times$ the enclosure width.

5 Numerical evidence

5.1 Three stages

Stage 1 (proof of concept). At $T = 100$, $d = 50$, the Gaussian window at $\sigma = L/4$ gives $H/d = 82.0\%$, already above the ceiling.

Stage 2 (window opposition). The literal Anthropic window $\phi(u) = \cos(\sqrt{2}|u|/L)^{1/2} \mathbf{1}_{|u| < L/2}$ has a jump discontinuity at its boundary, hence a $\hat{\phi}$ tail $\sim 0.7/t$ (a sinc tail), which produces hundreds of spurious oscillations and renders it unusable as a P2 signal. The two methods therefore *prefer opposite windows*: the inertia-based certificate wants a compact window with a sharp main lobe, while the eigenvector construction wants a smooth window with a fast-decaying $\hat{\phi}$. This opposition is structural evidence of complementarity, not a defect.

Stage 3 (optimal parameters). The bottleneck is not the window shape but the *atom overlap*. The Gaussian with $\sigma \geq 3L/4$ and d scaled to full coverage is the optimal configuration; among super-Gaussians, $p = 2$ (Gaussian) beats $p = 4, 6$, confirming that smoothness of $\hat{\phi}$ is the primary attribute.

5.2 The scaling artifact

An earlier run reported a decay of H/d to 0.627 at $T = 1000$, which would have invalidated the claim. We traced this to a window-inversion artifact: the atom positions τ_k were scaled with $\log(T)$ only, so the window failed to cover the zero support at large T . Once τ_d is scaled to cover $[0, 2T]$, the ratio stabilizes (Table 2). The correction is recorded explicitly because the artifact mimicked a genuine failure mode, and its diagnosis is part of the contribution.

6 Formal verification

The linear-algebra layer is machine-checked end-to-end.

Compiler. Lean 4.35.0-pre built from source in a network-restricted environment (stage0/stage1 bootstrap), with all eight Mathlib dependencies obtained through a repository-mirror pipeline that bypasses the blocked GitHub host.

Mathlib. 8139 modules compiled; the Gershgorin theorem `eigenvalue_mem_ball` is available.

Gershgorin. Theorem 2 is proven (not admitted): diagonal dominance implies $\det \neq 0$, and the eigenvalue lies in its disk. The collision identity $53 + 17 = 70$ is checked by `native_decide`.

Riemann–Siegel. The Z -function with the C_0 correction term is implemented in Lean (Init-only), with the first zero bracketed and its existence certified by the intermediate value theorem plus two sign checks.

Interval arithmetic. 30/30 zero enclosures verified; the artifact rejection is 15/15.

The dual-track verifier (Lean + SymPy) classifies every numerical claim as **green** (machine-proven), **yellow** (partially verified), or **red** (unverified and excluded from the main text).

7 Discussion and limitations

7.1 The honest ceiling

The central limitation is that Theorem 1 is numerical. The rigorous statement that would elevate it to a theorem is the following.

Conjecture (Winding identity). Let H be as constructed with $v_{\min} \in \ker(G)$. Then, for a finite Gabor matrix with the Toeplitz structure guaranteed by the Gabor Poisson summation (Shannon sampling), the number of real zeros of H in the support interval equals the positive index of $\tilde{G}$:

$$\#\text{zeros}(H) = n_+(\tilde{G}).$$

This is the finite-Gabor analogue of the Connes–van Suijlekom theorem, and we expect it to follow from the special Toeplitz/triangular form of the quadratic form in an orthonormal basis, together with the Hurwitz-theorem passage to the limit. We have not yet proven it.

7.2 The strongest counterargument

The sharpest objection to this note is: *a numerical ratio that decays in T is not a lower bound*. The correct statement is that the eigenvector route gives a strictly stronger *signal* than the moment route in the explored regime, but its promotion to an unconditional proportion requires (i) the winding identity of the Conjecture and (ii) a control of the prime-side error so that H 's zeros approximate the true zeros to $o(1/\log T)$. Until both are in place, the 87% figure is a guide to where the theorem should be, not the theorem itself.

A second objection: the Connes $x = 13$ numerical window cannot be transplanted directly, because at fixed T it corresponds to bandwidth $\lambda \rightarrow 0$ where the Anthropic certificate degenerates. We use only its *structure* (prolate-type minimal eigenvectors), never its numbers.

7.3 What is and is not proven

Proven (machine-checked): Gershgorin disk theorem; diagonal-dominance nonsingularity; the $53 + 17 = 70$ collision identity; the first 30 zero enclosures; the 15 false-positive rejections; the Riemann–Siegel Z -function at the first zero.

Numerical: $H/d \in [0.87, 0.94]$; matched-zero fraction ≈ 0.89 ; residual $\lesssim 10^{-15}$.

Open: the winding identity; the prime-side error bound; any unconditional proportion beyond $2/3$.

8 References

- [1] Anthropic (2026), *More Than Two Thirds of the Zeros of the Riemann Zeta Function Lie on the Critical Line*, with formalization and process note. anthropic.com/research/riemann-zeta
- [2] A. Connes (2026), *The Riemann Hypothesis: Past, Present and a Letter Through Time*, arXiv: 2602.04022.
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- [5] Baluyot, Goldston, Suriajaya, Turnage-Butterbaugh (2025), *Pair Correlation of Zeros of the Riemann Zeta Function I*, arXiv:2501.14545.
- [6] Guth, Maynard (2024), *New bounds for the Riemann zeta function on the critical line*, arXiv: 2405.20552.

Scope note. This note is an independent exploration. No claim herein is a claim about the Riemann Hypothesis itself: the method neither assumes nor refutes RH, and the collision is aimed at the *proportion constant* and at *constructive certificates*, following the scope of the Anthropic paper.